Introduction to Cisco Telemetry Broker (CTB)

Ajit Thyagarajan
Distinguished Speaker & Chief Architect

DEVNET-2340

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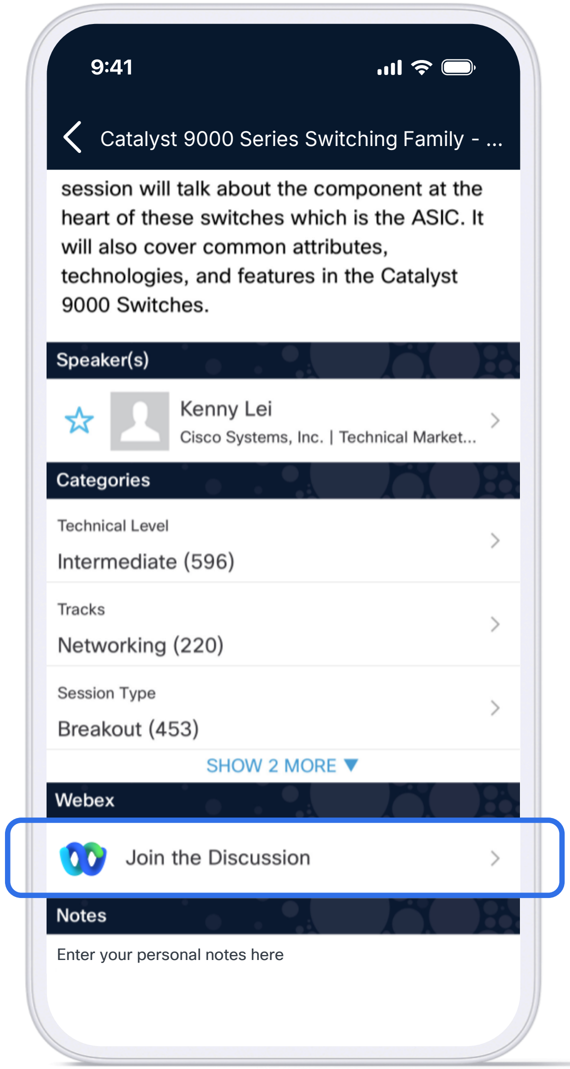

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#> whoami

- University of Delaware — Multicast, NTP
- Torrent Networking Technologies Corporation — High-speed Routing
- Ericsson — IGMP
- Fidelis Cybersecurity — Network Security
- AtomicMole — IoT, ML, AI
- Cisco — Telemetry, NDR

American Contract Bridge League

Ajit Thyagarajan
ajthyaga@cisco.com

Chief Security Architect
Distinguished Engineer
Cisco Secure Analytics

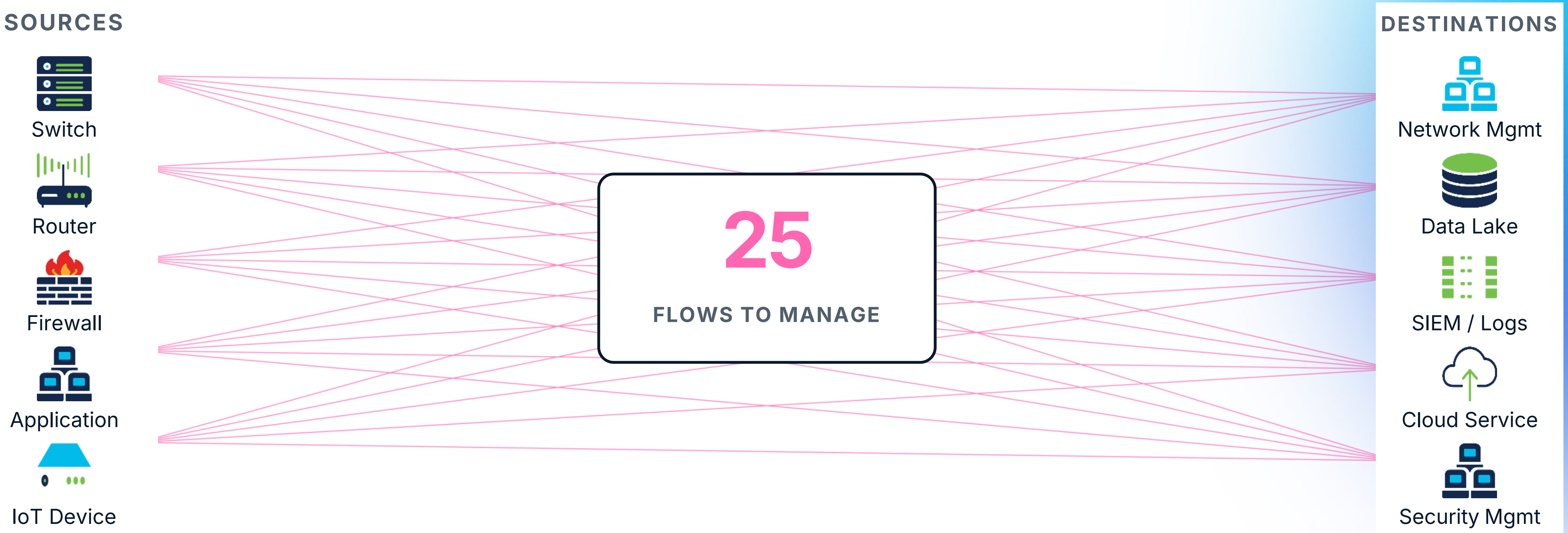
Agenda

1. The Telemetry Conundrum
2. Intelligent Telemetry Plane
3. Deployment
4. Demo
5. Learn more

The Telemetry Conundrum

Cisco Live!

BEFORE CTB

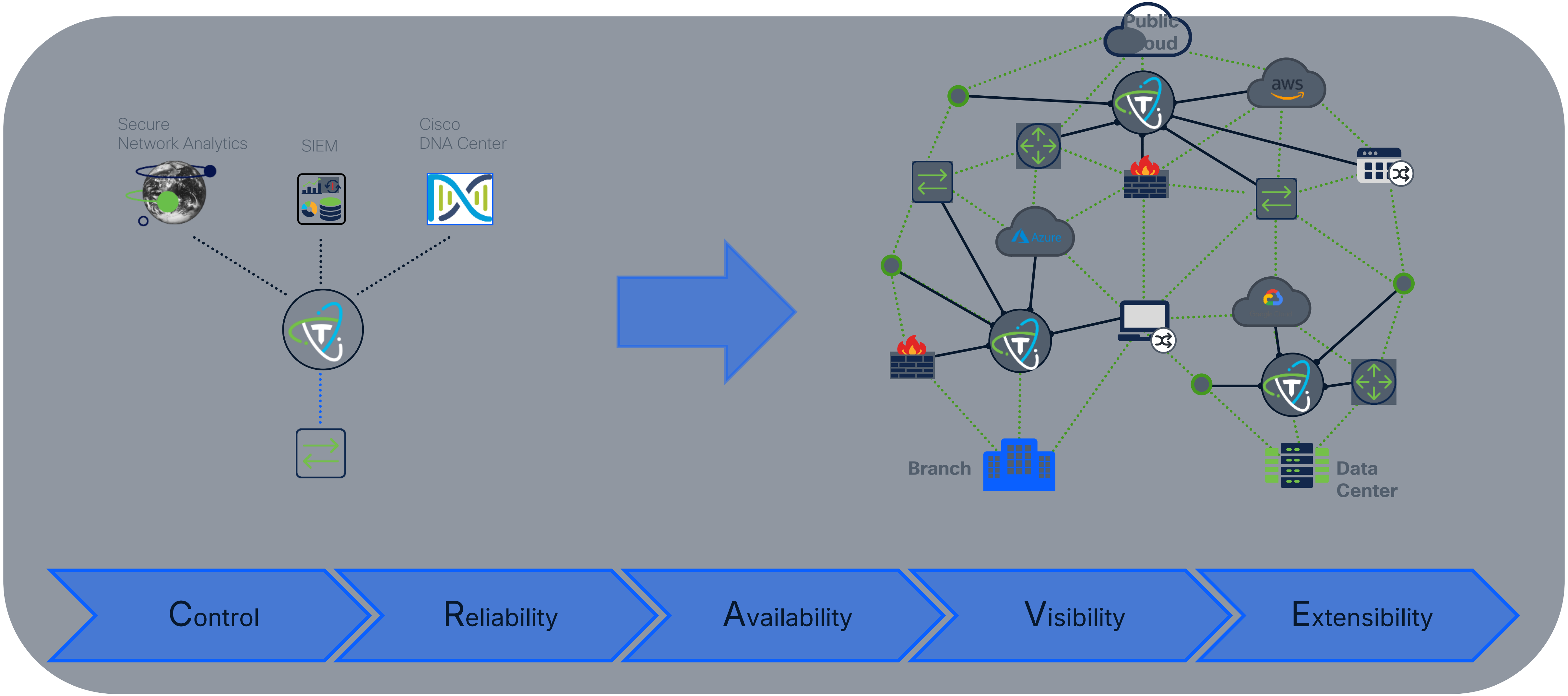


AFTER CTB


SOURCES
Switch
Router
Firewall
Application
IoT Device
Cisco Telemetry Broker
BROKER · FILTER · TRANSFORM
DESTINATIONS
Network Mgmt
Data Lake
SIEM / Logs
Cloud Service
Security Mgmt

Intelligent Telemetry Plane

Intelligent Telemetry Plane (ITP)



Cisco Telemetry Broker Democratizes Telemetry Data

Brokering

The ability to route and replicate telemetry data from a source location to multiple destinations

Filtering

The ability to filter data being replicated to enable fine grain control over what destinations ingest and analyze

Transforming

The ability to transform data protocols from the source to the destinations protocol of choice

Brokering

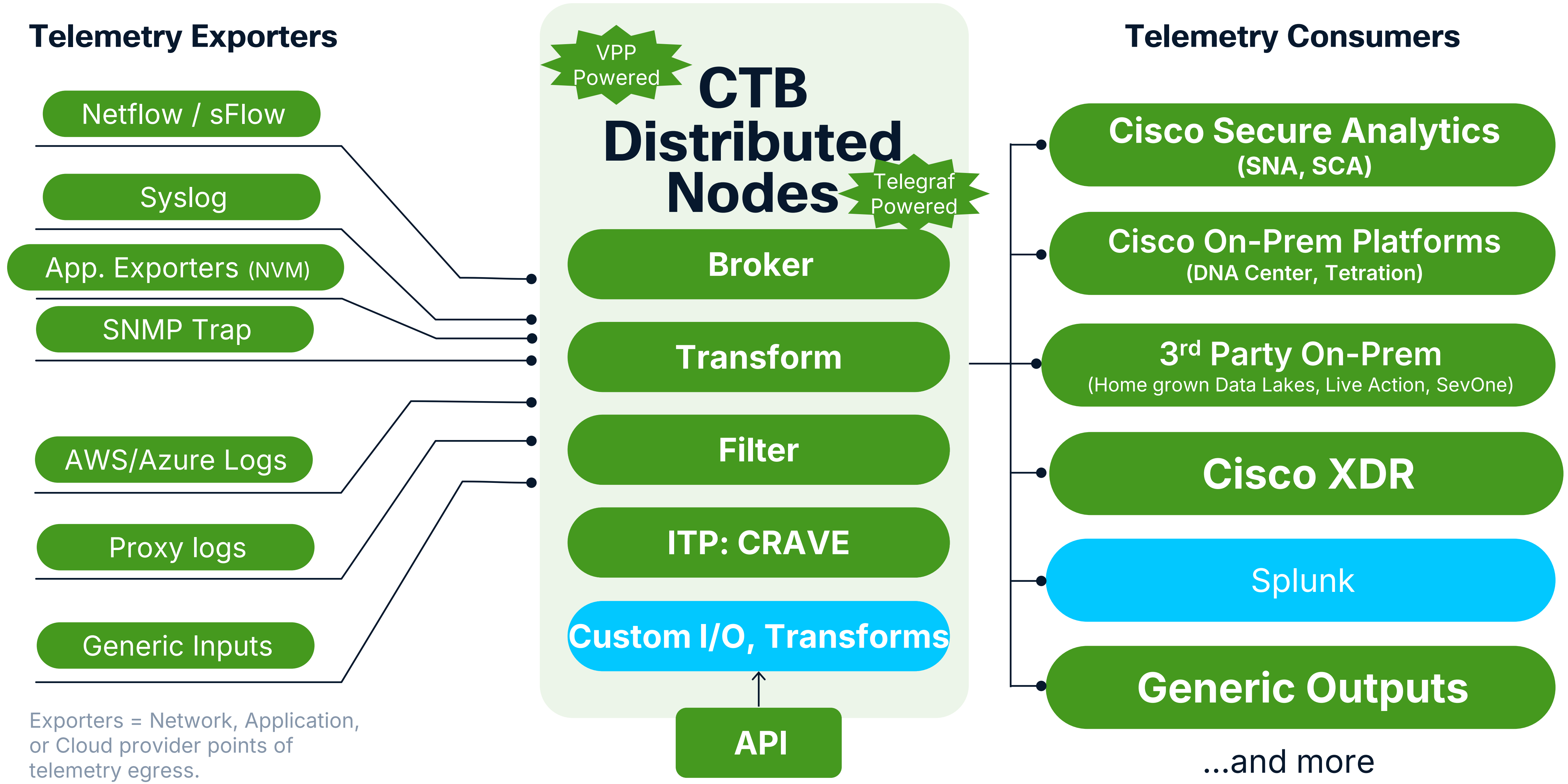
- Replicate Telemetry (20x perf improvement over UDP Director) >>>>>> Quickly enable PoV/onboarding of non-incumbent tools
- Route Telemetry (Cloud, On Prem, Container, Serverless) >>>>>> Let teams run the tools of their choice without deploying new agents/collectors

Filtering

- Filter to Drop & Segment >>>>>>
  - Control Costs: Pay Splunk to only index high value data
  - Compliance: Keep low value data in low-cost storage

Transformation

- Protocol Transformation >>>>>> Increased visibility of legacy sources, and into cloud sources
- Data Transformation >>>>>> Aggregate data, transform fields, add context
- Anonymize PII >>>>>> GDPR compliance to all downstream tools



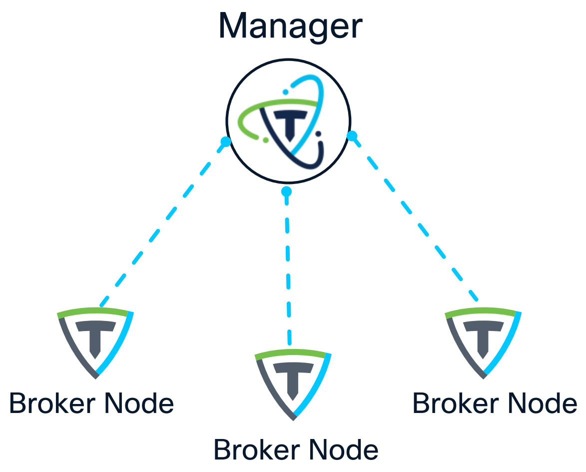
Exporters = Network, Application, or Cloud provider points of telemetry egress.

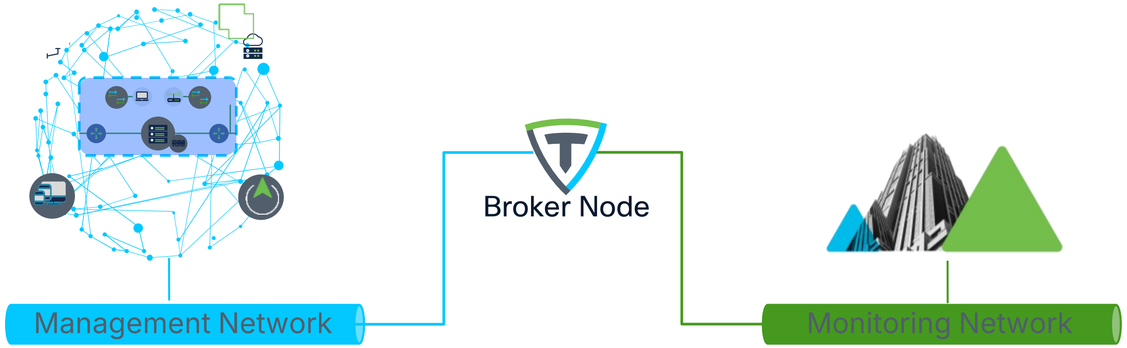
Available now
Coming Soon

Deployment

CTB Components

- CTB Manager node:
  - Only one manager is deployed and can manage multiple Broker nodes*
  - Maintains the policy/rules for the broker nodes enabling central management from one view
  - If the manager goes down, broker nodes continue to process telemetry
  - Backup configurations are created for recovery
- CTB Broker node:
  - Where the telemetry brokering work occurs
  - Can be deployed closest to telemetry sources





Deployment specifications

Minimum requirements for a CTB Manager:

- CPU: 4 cores
- Memory: 8 GB
- Storage: 100GB

vmware vSphere

Versions 7.0, 6.7 or 6.5

KVM

NUTANIX

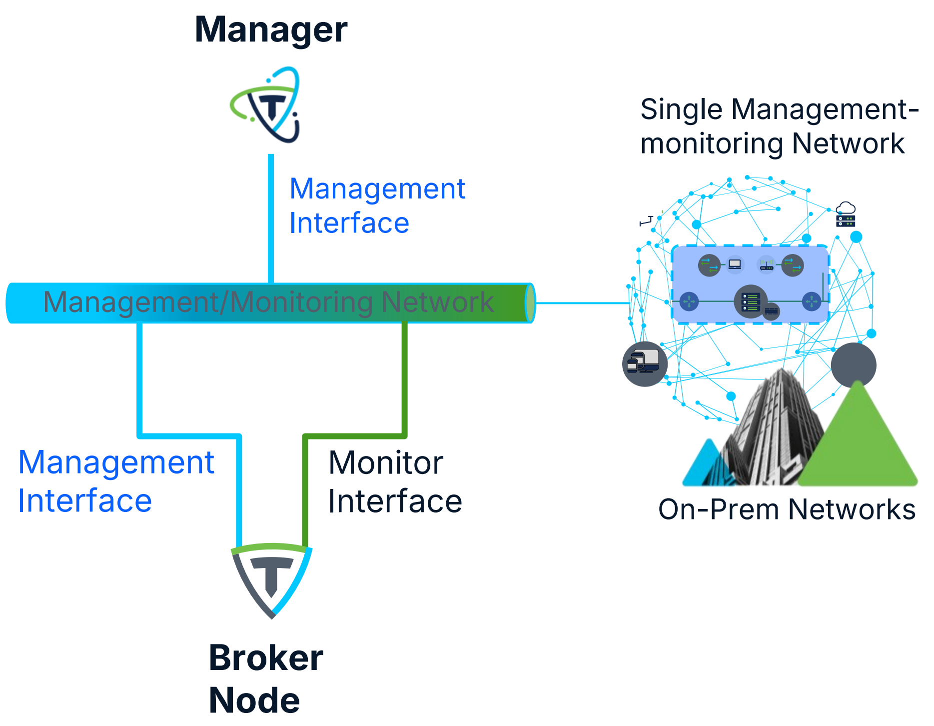
Minimum requirements for a CTB Broker Node:

- CPU: 8 cores
- Memory: 12 GB RAM
- Storage: 60 GB

TB2300 now available (Broker Node only)
(TB2400 available FY27)

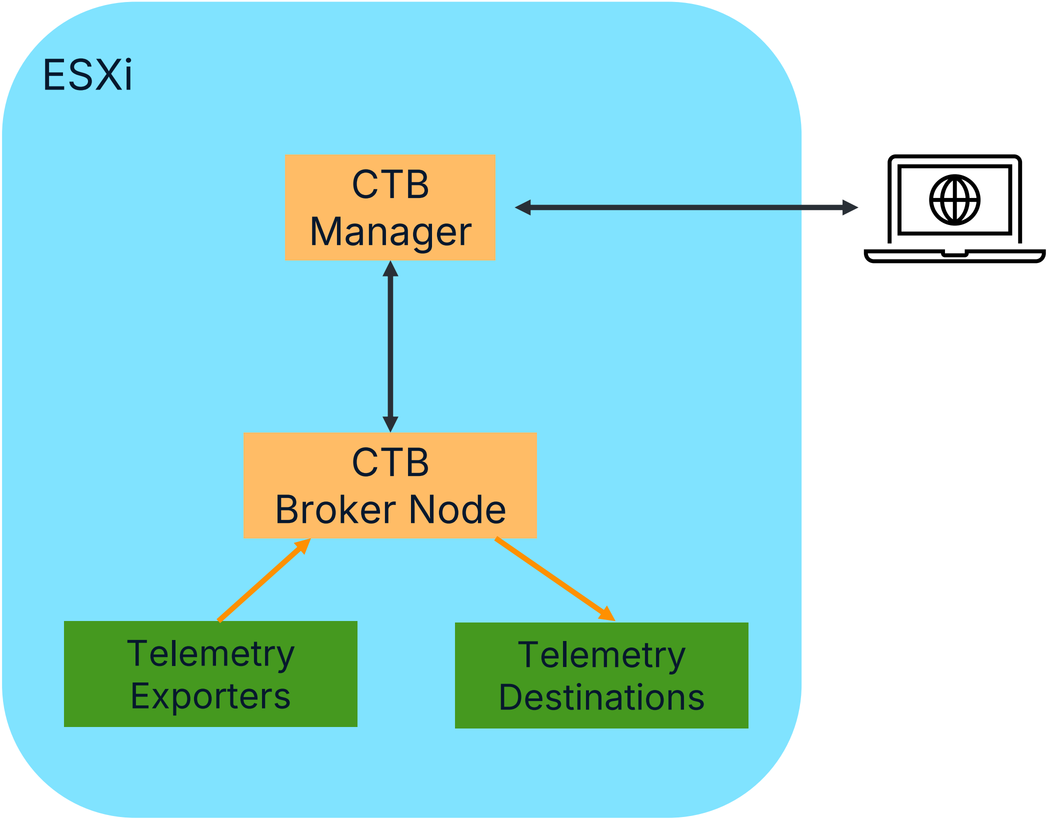
CTB Network Design (Basic)

- The Manager node has a single interface
  - Dedicated for management traffic only
- Broker nodes have two virtual interfaces
  - One dedicated for management
  - One dedicated for monitoring UDP traffic
- For smaller deployments that do not have a segmented management network all exporters of telemetry are sent to the Broker Node on a single management/monitoring network
- The Broker Node must have two IPs (v4 or v6) for Management and Monitoring interfaces (can be on same subnet)



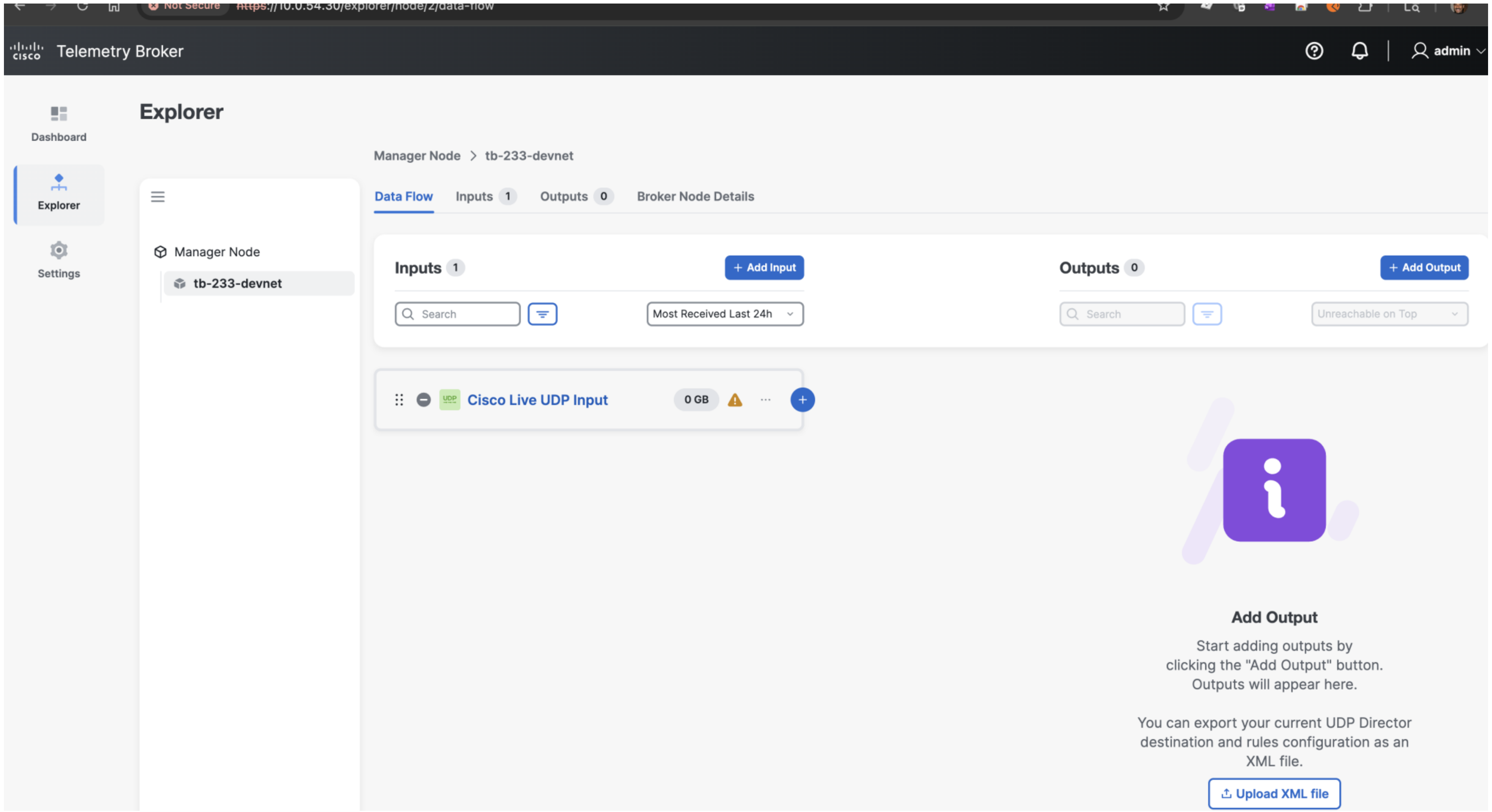
CTB Demo

Demo Setup

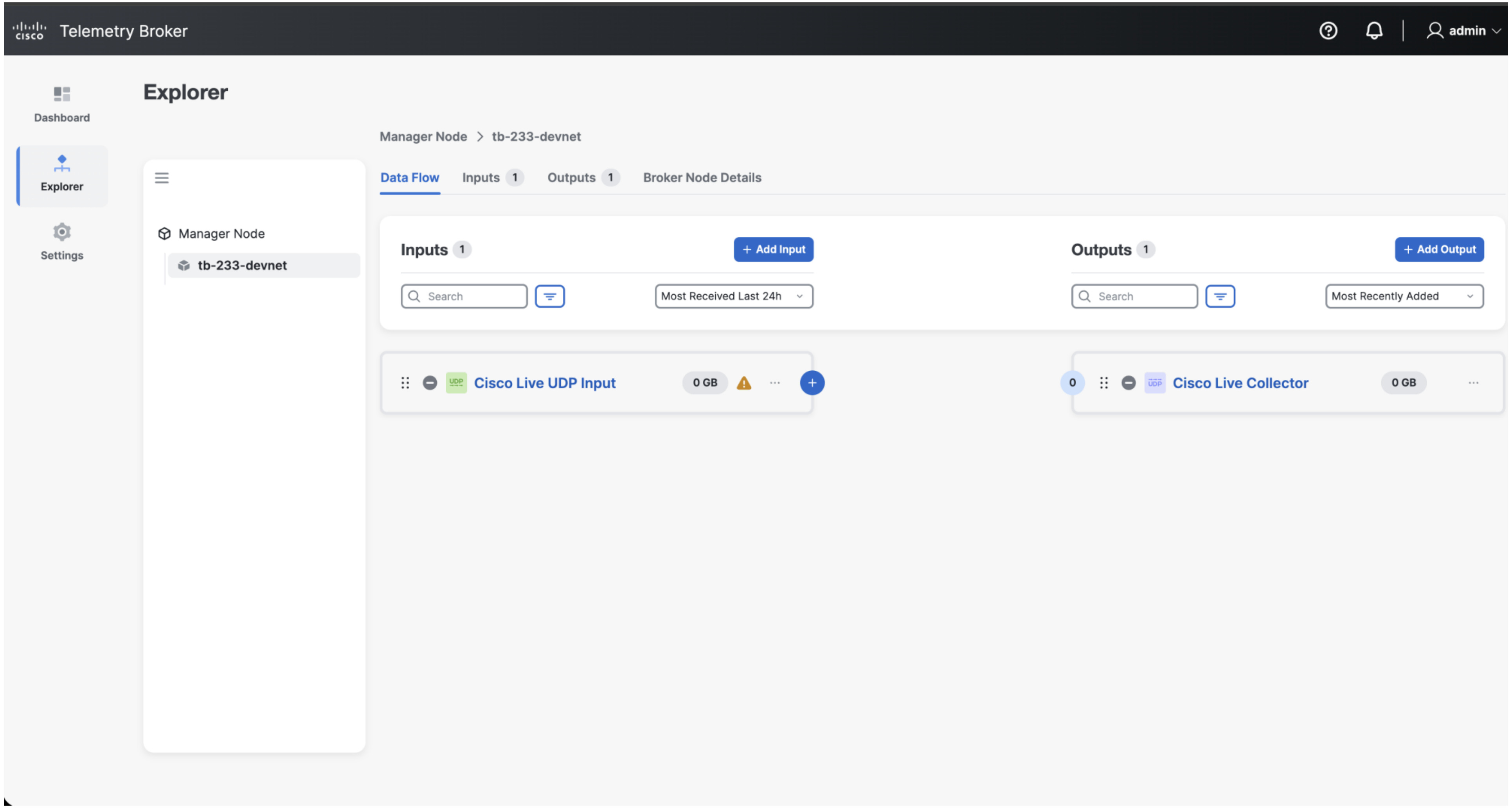


- 1 Broker Node has been registered to the Manager
- 1 Telemetry exporter has been configured to send IPFIX data to Broker Node's Telemetry Interface

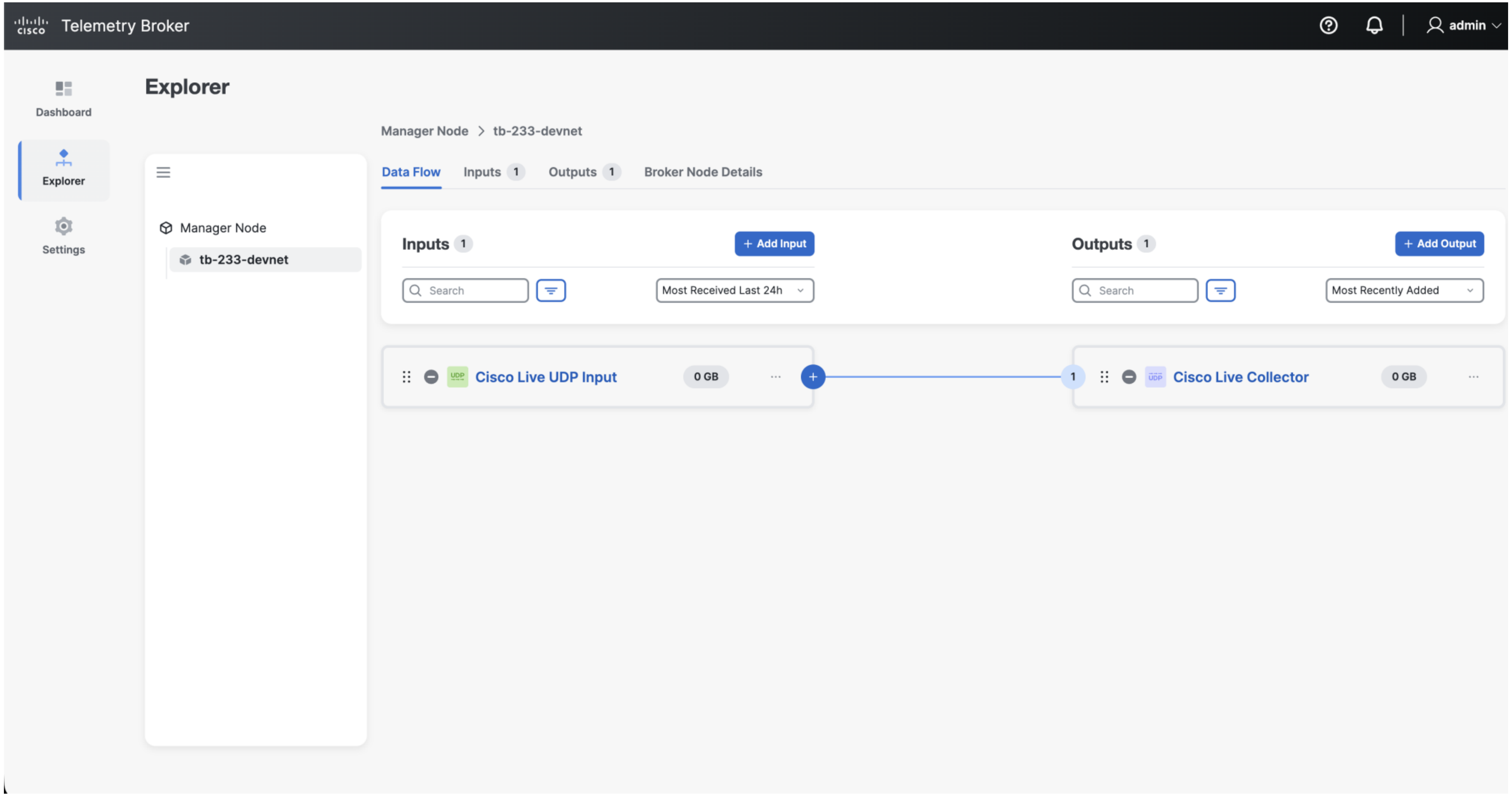
Demo 1.1: Configure Inputs



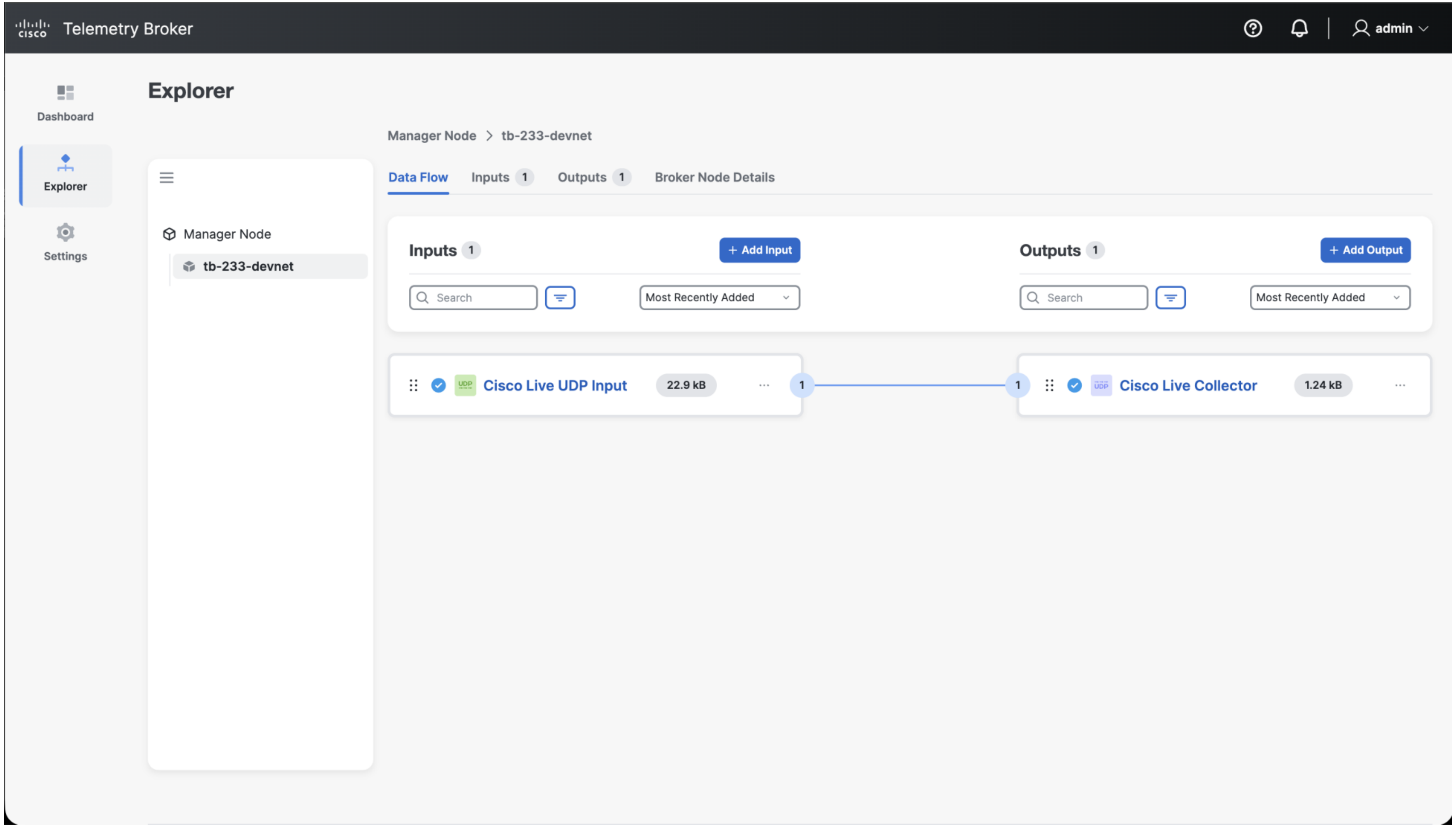
Demo 1.2: Configure Destinations



Demo 1.3: Configure Rules



Demo 1.4: Visualize Data



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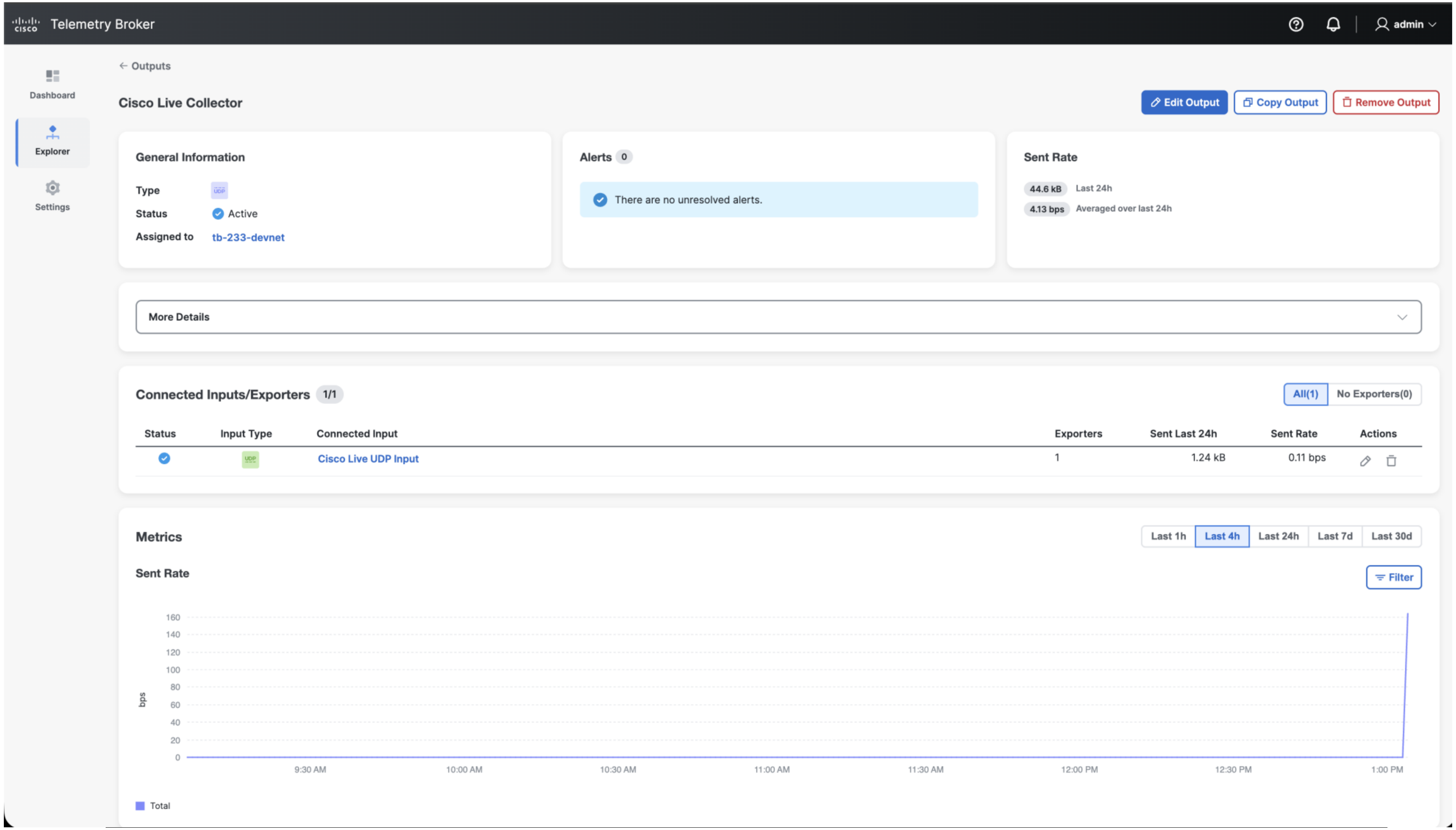
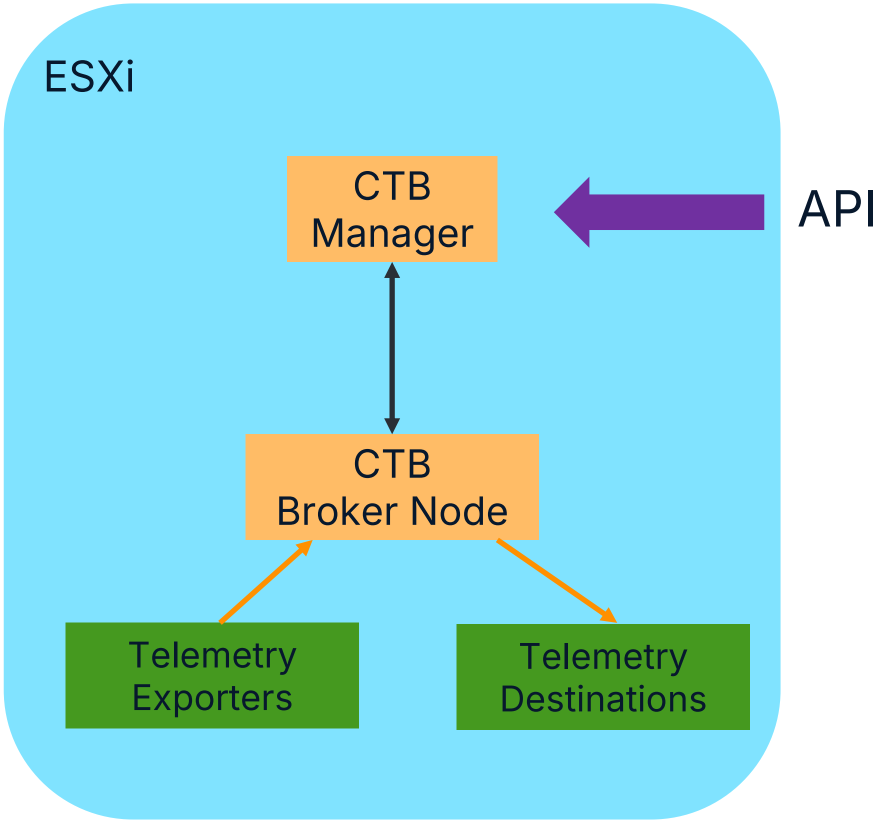


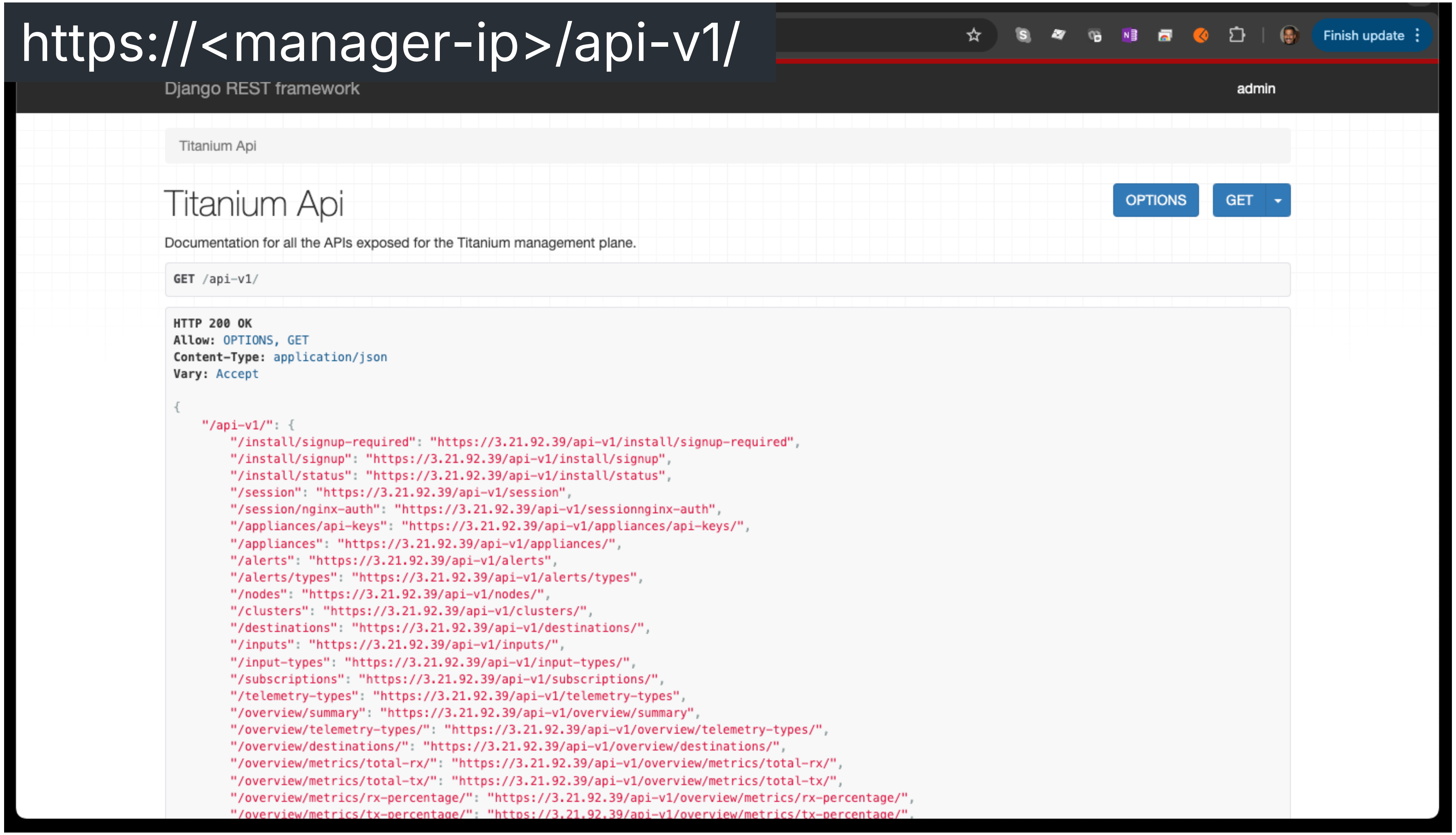
CTB API Demo

API Demo Setup

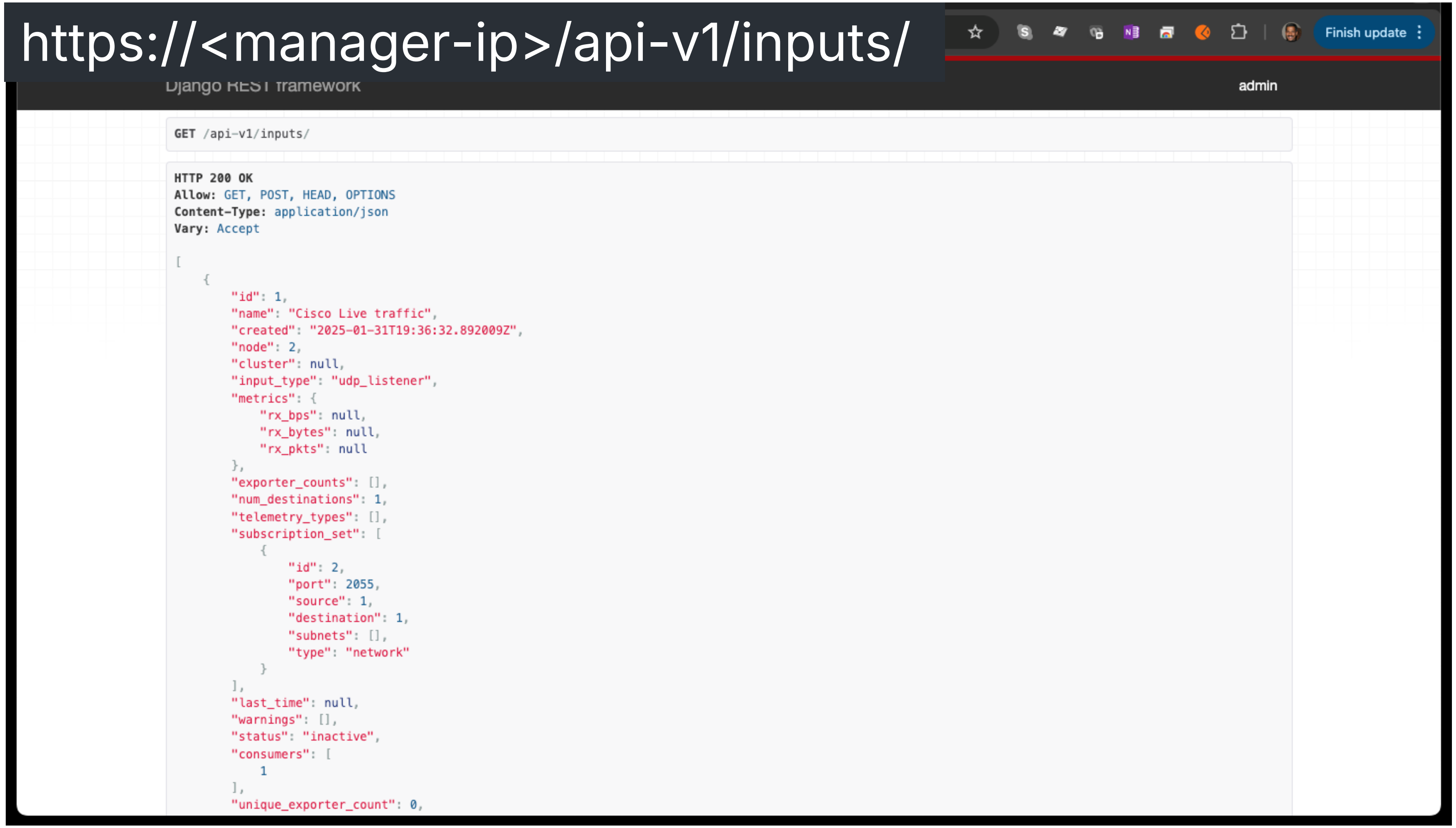


- CTB Manager exposes comprehensive API
- Can control all aspects of CTB via API
- Broker Nodes controlled via Manager API
- Retrieve all operational statistics via API

Demo 2.1: CTB Manager API



Demo 2.1: CTB Manager API



CTB MCP Server

Learn More

CTB Resources

- Learn more about CTB at https://cs.co/telemetrybroker
- Download CTB images at https://software.cisco.com (will need to register)
  - Deployment and User Guides have all the relevant information on how to operate CTB
- PDF of the Session will be available in the Mobile app as soon as this session is over
- If you want to try out the APIs on your CTB install, here is a link to how you can do it: https://github.com/brookmoor/clus2026-devnet-2340
- Use Webex teams room for this session to ask any questions:
  - DEVNET-2340 Introducing the Cisco Telemetry Broker (CTB)
- Reach out to me with any questions or suggestions (mailto: ajthyaga@cisco.com)

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